

ANNIHILATOR-POWER DYNAMICS ON IDEALS OF RESIDUE RINGS: RESONANT FIXED POINTS, TWO-CYCLES, AND EXACT TRANSIENTS

ANONYMOUS

ABSTRACT. Fix $r \geq 2$. On the ideal lattice of $\mathbb{Z}/N\mathbb{Z}$ we study $T_r(I) = \text{Ann}(I)^r$. For a prime-power component p^a , the valuation coordinate is the clipped reflection

$$f_{a,r}(e) = \min\{a, r(a - e)\}, \quad 0 \leq e \leq a.$$

The endpoints form a two-cycle, while an interior fixed point exists exactly when $r + 1$ divides a . Before clipping, the centered deviation $\Delta(e) = (r + 1)e - ra$ is multiplied by $-r$ at every step. This gives an exact hitting-time formula, a closed transient cumulative distribution, and a sharp logarithmic maximum depth. Chinese remaindering then yields the complete recurrent set, all fixed and two-cycle counts, and the Artin–Mazur zeta function for arbitrary N . Literal ideal arithmetic for every $N \leq 1000$ independently checks the valuation proof.

1. MAP AND OWNERSHIP BOUNDARY

Let $N \geq 2$, $R_N = \mathbb{Z}/N\mathbb{Z}$, and let $\mathcal{I}_N$ be its finite set of ideals. For a fixed integer $r \geq 2$, define

$$(1) \quad T_r: \mathcal{I}_N \longrightarrow \mathcal{I}_N, \quad T_r(I) = \text{Ann}_{R_N}(I)^r.$$

Ideal annihilators, ideal powers, and the Chinese remainder theorem are standard commutative-algebra ingredients; see Atiyah and Macdonald [3]. Annihilating-ideal graphs and their residue-ring specializations are also an established topic [1, 4]. We assign these ingredients no novelty credit. More directly, Schwiebert treats the radical and annihilator as operators on the ideal poset, studies their generated monoid, and analyzes finite-product behavior [5]. That operator-dynamics viewpoint, the bare annihilator iteration, and its product-decomposition framework are owned background here. Our bounded target is only the temporal package for the specific power-after-annihilator self-map (1): the clipped-reflection normal form, the divisibility-controlled fixed-point anomaly, exact transient layers, and the full one/two-cycle census. A targeted search completed on 29 August 2026 did not locate that conjunction. This is not an exhaustive owner certificate; external circulation and every priority claim remain on **HOLD**.

Factor N as

$$(2) \quad N = \prod_{i=1}^s p_i^{a_i}.$$

The Chinese remainder theorem identifies $\mathcal{I}_N$ with $\prod_i \{0, 1, \dots, a_i\}$: the coordinate e_i denotes the ideal $(p_i^{e_i})$ in $\mathbb{Z}/p_i^{a_i}\mathbb{Z}$. The first proof route works entirely in these valuation coordinates. The control route instead represents an ideal of R_N by its unique divisor generator (d) , computes

$$(3) \quad T_r((d)) = (\gcd(N, (N/d)^r)),$$

and only then compares with the coordinate formulas.

Date: Internal Stage 2 draft, 29 August 2026.

2. PRIME-POWER NORMAL FORM

Fix $a \geq 1$ and abbreviate $f = f_{a,r}$. Since $\text{Ann}(p^e) = (p^{a-e})$ and $(p^b)^r = (p^{\min(a,rb)})$, we have the following exact model.

Proposition 2.1 (Clipped reflection). *On $\mathcal{I}_{p^a}$, the map (1) is conjugate to*

$$(4) \quad f(e) = \min\{a, r(a-e)\}, \quad 0 \leq e \leq a.$$

Put $\Delta(e) = (r+1)e - ra$. Whenever $f(e) < a$,

$$(5) \quad \Delta(f(e)) = -r\Delta(e).$$

Clipping occurs precisely when $\Delta(e) \leq -a/r$.

Proof. Only the deviation statements require verification. In the unclipped branch $f(e) = r(a-e)$, whence

$$(r+1)f(e) - ra = r((r+1)(a-e) - a) = -r\Delta(e).$$

The inequality $r(a-e) \geq a$ is equivalent to $(r+1)e - ra \leq -a/r$. $\square$

The endpoints satisfy $0 \leftrightarrow a$. An interior point is fixed if and only if $e = r(a-e)$, which already exposes the arithmetic anomaly.

Theorem 2.2 (Recurrent states and resonance). *The recurrent set of $f_{a,r}$ is*

$$(6) \quad \{0, a\} \cup \left\{ \frac{ra}{r+1} : (r+1) \mid a \right\}.$$

The endpoints form one two-cycle. The displayed interior state, when it exists, is fixed. There are no other cycles.

Proof. The endpoint statements and the fixed-point equation follow from (4). Suppose $\Delta(e) \neq 0$ and the orbit has not clipped. By (5), its deviation alternates sign and its absolute value is multiplied by $r > 1$. It therefore eventually enters the clipping region. The next state is a , hence belongs to the endpoint two-cycle. Thus every nonfixed interior state is transient. $\square$

3. EXACT TRANSIENT LAYERS

For a state e , let $\text{depth}(e)$ be the first time its orbit is recurrent. The deviation gives a closed hitting-time clock rather than only an upper bound.

Theorem 3.1 (Exact depth). *For e in (6), $\text{depth}(e) = 0$. For every other state, put $D = \Delta(e)$. If $D < 0$, let*

$$k_- = \min\{k \geq 0 : r^{2k+1}|D| \geq a\};$$

if $D > 0$, let

$$k_+ = \min\{k \geq 0 : r^{2k+2}D \geq a\}.$$

Then

$$(7) \quad \text{depth}(e) = \begin{cases} 2k_- + 1, & D < 0, \\ 2k_+ + 2, & D > 0. \end{cases}$$

Consequently, on each sign side that contains transient states, the largest depth occurs at the smallest available value of $|D|$. The global maximum is the larger of those sidewise values, or zero when there are no transient states. In particular it is $O(\log_r a)$, and (7) is a sharp formula for every (a, r) , including $a = 1$.

Proof. Before clipping, the deviation at time j is $(-r)^j D$. A negative deviation clips on the following update exactly when its magnitude times r is at least a . If $D < 0$, negative signs occur at times $2k$; if $D > 0$, they occur at times $2k+1$. The two displayed threshold conditions and the additional clipping update give (7). Monotonicity in $|D|$ proves the sharp maximum statement. $\square$

For completeness, the whole transient distribution also has a floor/ceiling form. Put $\varepsilon_{a,r} = \mathbf{1}_{(r+1)|a}$ and, for $L \geq 1$,

$$(8) \quad M_-(L) = \max \left\{ 0, \min \left(a - 1, \left\lfloor \frac{ra - L}{r + 1} \right\rfloor \right) \right\},$$

$$(9) \quad M_+(L) = \max \left\{ 0, a - \max \left(1, \left\lceil \frac{ra + L}{r + 1} \right\rceil \right) \right\}.$$

These count interior states with $\Delta \leq -L$ and $\Delta \geq L$, respectively.

Corollary 3.2 (Transient cumulative distribution). *Let $C_{a,r}(t) = \#\{e : \text{depth}(e) \leq t\}$. Then $C_{a,r}(0) = 2 + \varepsilon_{a,r}$. For $t \geq 1$, set*

$$j_-(t) = 2 \left\lfloor \frac{t-1}{2} \right\rfloor, \quad L_-(t) = \left\lceil \frac{a}{r^{j_-(t)+1}} \right\rceil.$$

For $t \geq 2$, also set

$$j_+(t) = 2 \left\lfloor \frac{t-2}{2} \right\rfloor + 1, \quad L_+(t) = \left\lceil \frac{a}{r^{j_+(t)+1}} \right\rceil.$$

Then

$$(10) \quad C_{a,r}(t) = 2 + \varepsilon_{a,r} + M_-(L_-(t)) + \mathbf{1}_{t \geq 2} M_+(L_+(t)).$$

Thus the exact depth- t shell is $C_{a,r}(t) - C_{a,r}(t-1)$.

Proof. Among negative starting deviations, the largest even exponent available by time $t-1$ is $j_-(t)$; among positive ones, the largest admissible odd exponent is $j_+(t)$, which exists only for $t \geq 2$. The clipping test in [proposition 2.1](#) becomes $|D| \geq L_-(t)$ or $D \geq L_+(t)$. Counting the arithmetic progression $D = (r+1)e - ra$ over $1 \leq e \leq a-1$ gives (8)–(9). The endpoints and possible resonant fixed point contribute $2 + \varepsilon_{a,r}$. $\square$

4. CHINESE PRODUCTS, CYCLES, AND ZETA

Return to (2) and put

$$(11) \quad A = \prod_{i=1}^s \varepsilon_{a_i,r}, \quad B = \prod_{i=1}^s (2 + \varepsilon_{a_i,r}).$$

Theorem 4.1 (Complete finite dynamics). *Under Chinese remaindering, T_r is the product of the maps $f_{a_i,r}$. Its recurrent set has size B , its maximum transient depth is*

$$(12) \quad \max_{1 \leq i \leq s} \max_{0 \leq e \leq a_i} \text{depth}_{a_i,r}(e),$$

and the number of states of depth at most t is

$$(13) \quad \prod_{i=1}^s C_{a_i,r}(t).$$

For every $k \geq 1$,

$$(14) \quad \# \text{Fix}(T_r^k) = \begin{cases} A, & k \text{ odd}, \\ B, & k \text{ even}. \end{cases}$$

Hence there are A fixed points and $(B-A)/2$ two-cycles, and the Artin–Mazur zeta function [2] is

$$(15) \quad \zeta_{T_r}(z) = (1-z)^{-A} (1-z^2)^{-(B-A)/2}.$$

No period greater than two occurs.

Proof. Annihilators, products, and powers commute with the finite product decomposition of R_N , giving the coordinate product. A product point is recurrent exactly when each coordinate is recurrent; every such coordinate has period one or two, so the product period also divides two. Depth is the maximum coordinate depth, proving (12), while independent coordinate choices give (13). An odd iterate fixes only the resonant fixed coordinate in every component, producing A . An even iterate fixes every recurrent coordinate, producing B . Subtracting the fixed points from the points fixed by the square and dividing by two gives the cycle counts and (15). $\square$

The dynamics depends on the prime exponents and not on the primes themselves: moduli with the same exponent multiset have conjugate systems. This is a nonrigidity statement, not a claim that the phase portrait always recovers that multiset.

The internal collision boundary is also explicit. P100 uses least-valuation digit erasure on residue classes and reconstructs prime data from a transient profile; it does not act on the ideal lattice or alternate an annihilator with an ideal power. P102 uses an involution–norm map in a cyclic group algebra; its orbit structure comes from polynomial factorization rather than the clipped expanding reflection (4). Shared valuation, CRT, and zeta bookkeeping receive no contribution credit.

5. INDEPENDENT EXACT CONTROL

The accompanying standard-library verifier implements two independent lanes. The first enumerates $e = 0, \dots, a$, checks the deviation clock, the cumulative formula, and all iterate-fixed counts. The second enumerates actual divisor ideals (d) for every $N \leq 1000$, applies the literal gcd rule (3), and compares trajectories, depths, recurrent sets, and fixed counts with the CRT prediction. These finite computations are falsification controls, not proofs of the quantified theorems.

6. CONCLUSION

Annihilation followed by an ideal power becomes a clipped expanding reflection on each prime exponent. Its sign alternation forces a universal two-cycle, while the single equation $(r + 1)e = ra$ creates a sharp divisibility resonance. The same normal form supplies every transient layer and, after Chinese remaindering, the complete cycle and zeta data. Specialist owner review remains necessary before any external use.

REFERENCES

- [1] Mojgan Afkhami, Nesa Hoseini, and Kazem Khashyarmanesh. The annihilator ideal graph of a commutative ring. *Note di Matematica*, 36(1):1–17, 2016. doi: 10.1285/i15900932v36n1p1.
- [2] Michael Artin and Barry Mazur. On periodic points. *Annals of Mathematics*, 81(1):82–99, 1965. doi: 10.2307/1970384.
- [3] Michael F. Atiyah and Ian G. Macdonald. *Introduction to Commutative Algebra*. Addison–Wesley, Reading, Massachusetts, 1969.
- [4] M. Behboodi and Z. Rakeei. The annihilating-ideal graph of commutative rings I. *Journal of Algebra and Its Applications*, 10(4):727–739, 2011. doi: 10.1142/S0219498811004896.
- [5] Ryan C. Schwiebert. The radical-annihilator monoid of a ring. *Communications in Algebra*, 45(4):1601–1617, 2017. doi: 10.1080/00927872.2016.1222401.